

Math question bank — Grade 8–9

63 questions · 7 topics × (3 easy + 3 medium + 3 challenge) · For review before database seeding · Source: reviewed_generated

1. Linear Equations

Easy

#	Question	Answer
1	Solve: $2x + 3 = 11$	$x = 4$
2	Solve: $5x - 7 = 18$	$x = 5$
3	Solve: $x/3 + 2 = 7$	$x = 15$

Medium

#	Question	Answer
4	Solve: $3(x + 4) = 2x + 20$	$x = 8$
5	Solve: $(2x - 1)/3 = 5$	$x = 8$
6	Two numbers differ by 6 and add up to 40. Find the numbers.	17 and 23

Challenge

#	Question	Answer
7	A two-digit number has digits that add up to 9. Reversing its digits increases the number by 27. Find the number.	36
8	Solve: $(x + 2)/4 - (x - 1)/6 = 1$	$x = 4$
9	A is twice as old as B. Five years ago, A was three times as old as B. Find their present ages.	$A = 20, B = 10$

2. Exponents & Powers

Easy

#	Question	Answer
10	Find the value of 2^5 .	32
11	Simplify: $3^2 \times 3^3$	$3^5 = 243$
12	What is the value of 10^0 ?	1

Medium

#	Question	Answer
13	Simplify: $(2^3)^2$	64

14	Find the value of 5^{-2} .	$1/25$
15	Express 0.00042 in standard form.	4.2×10^{-4}

Challenge

#	Question	Answer
16	Simplify: $(2^4 \times 2^6) \div 2^8$	4
17	If $2^x = 64$, find x.	$x = 6$
18	Find the value of $(3/4)^{-2}$.	$16/9$

3. Polynomials & Factoring

Easy

#	Question	Answer
19	Add: $(3x + 5) + (2x - 3)$	$5x + 2$
20	Multiply: $4x \times 3x^2$	$12x^3$
21	Factorise: $6x + 9$	$3(2x + 3)$

Medium

#	Question	Answer
22	Expand: $(x + 3)(x + 5)$	$x^2 + 8x + 15$
23	Factorise: $x^2 - 25$	$(x - 5)(x + 5)$
24	Factorise: $x^2 + 7x + 12$	$(x + 3)(x + 4)$

Challenge

#	Question	Answer
25	Factorise: $2x^2 + 7x + 3$	$(2x + 1)(x + 3)$
26	If $x + 1/x = 4$, find the value of $x^2 + 1/x^2$.	14
27	Expand and simplify: $(x + 2)^2 - (x - 2)^2$	$8x$

4. Triangles & Circles

Easy

#	Question	Answer
28	Two angles of a triangle are 65° and 45° . Find the third angle.	70°
29	A right triangle has legs 3 cm and 4 cm. Find the hypotenuse.	5 cm
30	What is the measure of each angle of an equilateral triangle?	60°

Medium

#	Question	Answer
31	A right triangle has hypotenuse 13 cm and one leg 5 cm. Find the other leg.	12 cm
32	An exterior angle of a triangle is 110° and one interior opposite angle is 60° . Find the other interior opposite angle.	50°
33	The angles of a triangle are in the ratio 2 : 3 : 4. Find all three angles.	$40^\circ, 60^\circ, 80^\circ$

Challenge

#	Question	Answer
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34	A 10 m ladder leans against a wall with its foot 6 m from the wall. How high up the wall does it reach?	8 m
35	In an isosceles triangle the vertex angle is 40° . Find each base angle.	70°
36	What is the measure of an angle drawn in a semicircle (subtended by the diameter)?	90°

5. Surface Area & Volume

Easy

#	Question	Answer
37	Find the volume of a cube with side 4 cm.	64 cm ³
38	Find the volume of a cuboid 5 cm × 3 cm × 2 cm.	30 cm ³
39	Find the total surface area of a cube with side 3 cm.	54 cm ²

Medium

#	Question	Answer
40	Find the total surface area of a cuboid 4 cm × 3 cm × 2 cm.	52 cm ²
41	Find the volume of a cylinder with radius 7 cm and height 10 cm. (Use $\pi = 22/7$)	1540 cm ³
42	A cube has a volume of 125 cm ³ . Find the length of one side.	5 cm

Challenge

#	Question	Answer
43	How many cubes of side 2 cm can fit exactly inside a box measuring 6 cm × 4 cm × 2 cm?	6
44	Find the curved surface area of a cylinder with radius 7 cm and height 15 cm. (Use $\pi = 22/7$)	660 cm ²
45	A cuboid-shaped tank measures 2 m × 1.5 m × 1 m. How many litres of water can it hold? (1 m ³ = 1000 L)	3000 litres

6. Probability & Statistics

Easy

#	Question	Answer
46	A fair coin is tossed once. What is the probability of getting heads?	1/2
47	A die is rolled once. What is the probability of getting an even number?	1/2
48	Find the mean of 5, 10, 15.	10

Medium

#	Question	Answer
49	A bag contains 3 red and 5 blue balls. One ball is drawn at random. What is the probability it is red?	3/8
50	Cards numbered 1 to 10 are shuffled and one is drawn. What is the probability of drawing a multiple of 3?	3/10
51	Find the median of 12, 5, 8, 15, 10.	10

Challenge

#	Question	Answer
52	Two fair coins are tossed together. What is the probability of getting exactly one head?	$\frac{1}{2}$
53	The mean of 10 numbers is 20. If each number is increased by 5, what is the new mean?	25
54	Two dice are rolled together. What is the probability that the sum is 7?	$\frac{1}{6}$

7. Money Problems

Easy

#	Question	Answer
55	An item is bought for 200 and sold for 250. Find the profit.	50
56	Find the simple interest on 1000 at 5% per year for 2 years.	100
57	A 10% discount is given on a marked price of 450. How much does the customer pay?	405

Medium

#	Question	Answer
58	An item bought for 400 is sold for 460. Find the profit percentage.	15%
59	Find the simple interest on 2500 at 8% per year for 3 years.	600
60	An item is sold for 540 at a loss of 10%. Find the cost price.	600

Challenge

#	Question	Answer
61	Successive discounts of 20% and then 10% are given on a marked price of 500. Find the final price.	360
62	In how many years will 4000 grow to 5000 at 5% per year simple interest?	5 years
63	The cost price of 12 items equals the selling price of 10 items. Find the profit percentage.	20%